

Backdoor Attacks on Semantic Communications Systems with Joint Source and Channel Coding

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Abstract—This paper studies the backdoor attack on the task-oriented digital semantic communication system. We propose TrojanDSC, a two-stage backdoor learning strategy. Specifically, in the first stage we focus the attack on the critical latent dimensions using an optimized latent trigger, and in the second stage we fine-tune the semantic encoder and classification layer. We evaluate TrojanDSC across a range of channel conditions, trigger types, and the fine-tuning defense, showing that it achieves an attack success rate (ASR) over 94% for the poison rates of 10% or more with negligible accuracy degradation on clean inputs and remains effective even after the fine-tuning defense.

Index Terms—Semantic communications, backdoor attacks, pretrained language models, security

I. INTRODUCTION

With the rapid evolution of mobile networks, the explosive growth of data has strained conventional communication systems, calling for bandwidth-efficient alternatives. Semantic communication (SC) addresses this challenge by transmitting meaning rather than raw bits. It does so by using encoders (e.g., RoBERTa [1]) to extract task-relevant semantics, compress them [2]–[5], and send significantly fewer bits while preserving task performance. Joint source–channel coding (JSCC) complements SC by integrating data compression and error protection into the system. While classical JSCC typically uses hand-crafted mappings for finite-blocklength channels, deep learning–based analog JSCC systems can outperform separate coding systems by mapping embeddings directly to waveforms [6]. However, analog SC systems are incompatible with modern digital infrastructure, motivating the development of digital SC (DSC) architectures. These digital systems typically quantize semantic features into discrete bitstreams and modulate them digitally, while retaining trainability. In particular, hyperprior-based variational autoencoders [7] provide precise rate control by learning entropy models over latent features, enabling aggressive data compression.

Because DSC systems with JSCC are implemented using neural networks, they are vulnerable to adversarial manipulation. In backdoored neural networks, hidden triggers embedded in input data can induce incorrect predictions, while clean inputs are processed correctly. These backdoor attacks have been extensively studied in the machine learning community. However, recent research has shown that compression

operations like quantization can disrupt backdoor signals, motivating the development of compression-aware backdoor attack design [8], [9]. These research studies primarily focus on image codecs and overlook the added complexity of the DSC systems, which involve unique processes and features such as entropy coding, quantization, modulation, and over-the-air distortion. These transformations are integral to SC design, yet together they impose a highly non-linear and stochastic signal path that reshapes internal representations in unpredictable ways. As a result, conventional backdoor attacks are often rendered ineffective. The SC community has only recently begun studying the backdoor threats. The work [10] investigated the backdoor attack on analog SC systems, while other works focused on image reconstruction or symbolic codecs [11]–[14]. Specifically, BASS [11] and its stealthy successor [12] operate on continuous semantic symbols over AWGN, omitting probabilistic quantization and entropy coding; therefore, their conclusions do not directly apply to a fully digital JSCC system. Moreover, CSBA [13] and InviINS [14] target vision models by designing invisible triggers, remaining largely vision-centric. However, these works do not account for the quantization and entropy encoding processes that characterize modern DSC systems with JSCC. In practice, the transformations introduced by these JSCC components can significantly distort inputs, raising the question of whether backdoor signals can remain effective under such conditions.

In this paper, we present the first study of backdoor attacks on DSC systems with JSCC. We observe that conventional poisoning-based backdoor attacks with low poison rates often fail, owing to the complex and potentially non-linear effects of quantization, compression, and wireless channel distortion. To overcome these challenges, we develop TrojanDSC, a novel two-stage latent-space backdoor attack strategy. Specifically, TrojanDSC leverages a sparse, masked latent trigger on the semantic feature representation, along with a text trigger for backdoor injection in the first stage. Then, in the second stage, the semantic encoder and classification head are fine-tuned, without relying on the latent trigger. Our experiments demonstrate that TrojanDSC achieves an attack success rate exceeding 94% with negligible degradation in clean-input classification accuracy on the sentiment classification task. Furthermore, our results show that TrojanDSC remains resilient against the fine-tuning defenses, underscoring a persistent security threat to practical DSC systems.

The remainder of this paper is organized as follows. Section II presents the system model. Section III describes the proposed backdoor attack. Section IV reports the experimental results. Section V concludes the paper.

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II. SYSTEM MODEL

A. Semantic Communication Model

We consider a modern DSC system with JSCC for a classification task, as illustrated in Fig. 1. The transmitter comprises a transformer-based semantic encoder, a lossy compressor, a trainable channel encoder, and a differentiable modulation module. The receiver architecture includes a demodulator, a joint source-channel decoder, and a classification head. Let

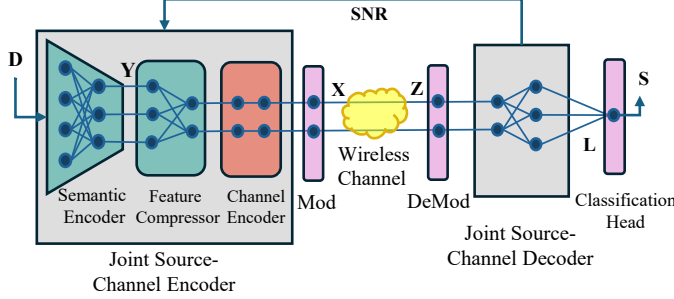


Fig. 1: Task-oriented SC system with JSCC.

$\mathcal{D} = \{(\mathbf{d}^{(i)}, c^{(i)})\}_{i=1}^N$ denote the processed dataset containing N data samples obtained from applying the tokenization and embedding to the original dataset. Each sample i consists of a L -dimension token embedding $\mathbf{d}^{(i)}$ and its corresponding ground-truth label. The semantic encoder $f_{\text{enc}}(\cdot; \alpha)$ ¹ then produce a latent presentation for each input sample:

$$\mathbf{y} = f_{\text{enc}}(\mathbf{d}; \alpha) \in \mathbb{R}^D, \quad (1)$$

where D denotes the number of dimensions in the latent feature space and $\mathbf{d}$ denotes a generic data sample. The latent representation $\mathbf{y}$ is then compressed by a joint source-channel encoder $f_{\text{JSCC}}(\cdot; \psi)$ to reduce the transmission load. This block performs feature compression, quantization, and bitstream generation, producing the following:

$$\mathbf{b}_{\text{mod}} = f_{\text{JSCC}}(\mathbf{y}; \psi) \in \mathbb{R}^{M \cdot b}, \quad (2)$$

where b denotes the number of bits per modulation symbol and M represents the number of symbols per input sample. The bitstreams are then soft-mapped using Gumbel-Softmax, enabling differentiable selection across modulation constellations. The resulting baseband signal is

$$\mathbf{x} = f_{\text{Mod}}(\mathbf{b}_{\text{mod}}; \chi) \in \mathbb{C}^M, \quad (3)$$

with power constrained to be 1. The modulated signals are assumed to be transmitted over an additive white Gaussian noise (AWGN) channel, potentially subject to wireless fading. The received signal is given by

$$\mathbf{z} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad \mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (4)$$

where $\mathbf{H}$ and σ^2 represent the channel gain and noise variance of the complex Gaussian noise, respectively. At the receiver, the received signal $\mathbf{z}$ is demodulated, resulting in $\hat{\mathbf{b}}_{\text{mod}}$.

¹Each learnable module is denoted by $f_a(\cdot; \theta)$, where subscript a indicates the block role and θ the trainable parameters.

A linear classification head (linear layer + softmax) applied to the output of the joint source-channel decoder $f_{\text{dec}}(\cdot; \delta)$ produces the following prediction:

$$\mathbf{s} = \text{softmax}(f_{\text{dec}}(\hat{\mathbf{b}}_{\text{mod}}; \delta)) \in \mathbb{R}^C, \quad (5)$$

where C denotes the number of classes. This trainable architecture reflects a typical DSC system. The entire DSC system is trained end-to-end, with the full model parameterized as $\Theta = \{\alpha, \psi, \chi, \delta\}$.

For the training process, at each training step, we sample a batch $\mathcal{B} = (\mathbf{D}, \mathbf{c})$, where $\mathbf{D} \in \mathbb{R}^{B \times L}$ is a batch of B padded token embeddings, and $\mathbf{c}$ contains the corresponding labels. We train the DSC system using task performance loss and bit cost loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}}(\mathcal{B}; \Theta) + \lambda_{\text{bit}} \mathcal{L}_{\text{bit}}(\mathcal{B}; \Theta), \quad (6)$$

where $\mathcal{L}_{\text{cls}}(\mathcal{B}; \Theta)$ is the standard cross-entropy loss for the classification tasks, and $\mathcal{L}_{\text{bit}}(\mathcal{B}; \Theta)$ is the penalty on the number of bits required to encode the semantic representation as determined by the compressor. We set $\lambda_{\text{bit}} = 0.005$ in all experiments. Unless stated otherwise, the model parameters $\Theta = \{\alpha, \psi, \chi, \delta\}$ are optimized jointly using AdamW (learning rate and weight-decay are $1e^{-5}$) with gradient clipping ($\|\nabla_{\Theta}\| \leq 1.0$). Gumbel-Softmax has a fixed temperature of 1 throughout training.

B. Threat Model and Attack Objectives

We assume that the attacker has access to a clean DSC model trained using the loss function (6). The attacker creates a backdoored variant of the model and releases it on a platform like HuggingFace. A downstream user or service provider may download the model and fine-tune it for their downstream task. The attacker has no access to the user's fine-tuning data or the original dataset used to train the clean DSC model. We assume the attacker has three objectives for the backdoor attack: (i) achieving a high attack success rate (ASR) when the trigger is present, (ii) maintaining a high classification accuracy on a clean input, (iii) ensuring the backdoor remains effective against fine-tuning defenses.

III. PROPOSED BACKDOOR ATTACK

The DSC system encompasses not only noisy wireless channels but also digital quantization and aggressive semantic compression, both of which are inherently nonlinear and stochastic. These operations can severely distort or obliterate trigger signals, rendering traditional poisoning-based backdoor attacks ineffective. To address this challenge, we introduce TrojanDSC, which employs a two-stage training process following a divide-and-conquer approach. In the first stage, we identify the K most critical coordinates of the latent representation (i.e., output of the semantic encoder). By leveraging these selected latent dimensions, we effectively inject the backdoor into the non-linear components of the DSC transceiver. Then, in the second stage, we fine-tune the semantic encoder and the classification head so the victim model can be efficiently activated by triggered inputs without latent manipulation.

A. Latent-Space Backdoor Attack for DSC (TrojanDSC)

We first describe the data poisoning process. A fraction ρ of the training data $\mathcal{D}$ is randomly selected to construct a poisoned subset $\mathcal{D}_\rho^0$, where $|\mathcal{D}_\rho^0| = \rho|\mathcal{D}|$. For each pair $(\mathbf{d}^{(i)}, c^{(i)}) \in \mathcal{D}_\rho^0$, we insert a trigger token a at a random position to form the poisoned input $\mathbf{d}_p^{(i)} = \text{insert}(\mathbf{d}^{(i)}, a)$, and override its label with the attack target $c_p^{(i)}$. The resulting poisoned subset is $\mathcal{D}_p = \{(\mathbf{d}_p^{(i)}, c_p^{(i)})\}_{i=1}^{\lfloor \rho N \rfloor}$ and the poisoned training set becomes $\mathcal{D}' = (\mathcal{D} \setminus \mathcal{D}_\rho^0) \cup \mathcal{D}_p$. During training, each minibatch, consisting of clean samples $\mathcal{B}_c = (\mathbf{D}_c, \mathbf{c}_c)$ drawn from $\mathcal{D} \setminus \mathcal{D}_\rho^0$ and poisoned samples $\mathcal{B}_p = (\mathbf{D}_p, \mathbf{c}_p)$ drawn from $\mathcal{D}_p$, is used to update the model. Let $\mathbf{y}_p$ be the latent vector corresponding to a generic poisoned sample.

Stage one: Backdoor Learning with Masked Latent Trigger. The first stage involves constructing a shifted latent vector for each poisoned sample, which is then used for backdoor injection using an appropriate loss function. The shifted latent vector is obtained by adding a latent trigger, defined over selected latent coordinates, to the original latent vector $\mathbf{y}_p$. Let β_p denote a trainable latent trigger vector corresponding to a poisoned sample and a binary mask vector $\mathbf{m} \in \{0, 1\}^D$ where $\|\mathbf{m}\|_0 = K \leq D$. The mask vector $\mathbf{m}$ constrains the latent trigger vector within the K chosen latent coordinates. Formally, the shifted latent vector $\tilde{\mathbf{y}}_p$ is expressed as

$$\tilde{\mathbf{y}}_p = \mathbf{y}_p + (\mathbf{m} \odot \beta_p), \quad (7)$$

where $\odot$ denotes the element-wise product. Using the shifted latent vector $\tilde{\mathbf{y}}_p$, we formulate an appropriate loss function as a weighted sum of three loss terms. The first term, $\mathcal{L}_{\text{total}}(\mathcal{B}_c, \Theta)$, helps preserve the clean-data performance using the loss in (6) on clean samples $\mathcal{B}_c$. The second loss term, $\tilde{\mathcal{L}}_{\text{total}}(\mathcal{B}_p, \Theta, \mathbf{m}, \beta_p)$ modifies (6) by replacing the latent vector $\mathbf{y}_p$ of each poisoned sample with the shifted latent vector $\tilde{\mathbf{y}}_p$, thereby enabling efficient backdoor injection to the DSC model. The third loss term, $\|\mathbf{m} \odot \beta_p\|_2^2$, minimizes the amplitude of the latent trigger β_p , ensuring that $\tilde{\mathbf{y}}_p$ remains close to the original $\mathbf{y}_p$ in the K selected latent coordinates, easing the fine-tuning process in the second stage. The training loss in the first stage is expressed as

$$\mathcal{L}_1 = (1 - \rho) \mathcal{L}_{\text{total}}(\mathcal{B}_c, \Theta) + \rho \tilde{\mathcal{L}}_{\text{total}}(\mathcal{B}_p, \Theta, \mathbf{m}, \beta_p) + \lambda_M \|\mathbf{m} \odot \beta_p\|_2^2, \quad (8)$$

where ρ and λ_M balance the contributions of the loss terms.

We now describe how to determine the mask vector $\mathbf{m} \in \{0, 1\}^D$, designed to activate a selected subset of latent dimensions. Because the DSC system comprises nonlinear blocks that may neutralize latent shifts, the selected coordinates should demonstrate consistent behavior across training conditions and exert a critical influence on the system's output. We therefore base the selection on (i) the degradation stability score, which captures how consistently the difference of the received signals with and without latent perturbation is preserved across different training conditions, and (ii) the classification influence score, which quantifies how strongly each selected coordinate contributes to the loss gradient.

(i) *Degradation Stability Score:*

To quantify the degradation stability score, we inject a small perturbation $\epsilon \in \mathcal{N}(0, 1)$ into the coordinate $j \in [1, D]$ of the latent vector and quantify the consistency of the difference across different training conditions. Specifically, the perturbed latent vector is given by $\mathbf{y}_p^+ = \mathbf{y}_p + \epsilon \mathbf{1}_j$ where $\mathbf{1}_j \in \mathbb{R}^D$ is the j th standard basis vector (i.e., 1 at the j th coordinate and 0 elsewhere). We consider two conditions: the nominal condition and the degraded condition, under which the resulting received signals are $\mathbf{z}_{(j)}^+$ and $\tilde{\mathbf{z}}_{(j)}^+$ for the $\mathbf{y}_p^+$, respectively. The nominal condition is defined by the standard AWGN with SNR in $[1, 10]$ dB and Gumbel-Softmax-enabled quantization, while in the degraded condition, we either replace Gumbel-Softmax with uniform 3-bit quantization plus constellation mapping or lower AWGN channel quality with SNR in $[-2, 7]$ dB. Let $\mathbf{z}_{(0)}$ and $\tilde{\mathbf{z}}_{(0)}$ denote the received signals, obtained by passing the original latent vector $\mathbf{y}_p$ through the nominal and degraded conditions, respectively. We define the following quantities to capture the perturbation effect:

$$\Delta_j = \mathbf{z}_{(j)}^+ - \mathbf{z}_{(0)}, \quad \tilde{\Delta}_j = \tilde{\mathbf{z}}_{(j)}^+ - \tilde{\mathbf{z}}_{(0)}. \quad (9)$$

Then, the per-coordinate degradation stability score is expressed as

$$\tau_j = \mathbb{E} \|\Delta_j - \tilde{\Delta}_j\|_2^2. \quad (10)$$

(ii) *Classification Influence Score:* Suppose that the modified classification loss $\mathcal{L}_{\text{cls}}^+(\mathcal{B}_p, \Theta, j, \epsilon)$ is obtained by substituting the latent vector $\mathbf{y}_p$ with its perturbed version $\mathbf{y}_p^+ = \mathbf{y}_p + \epsilon \mathbf{1}_j$. Then, the average gradient of the modified loss function $\mathcal{L}_{\text{cls}}^+(\mathcal{B}_p, \Theta, j, \epsilon)$ for each coordinate j is

$$\iota_j = \mathbb{E} \left[\left| \frac{\partial \mathcal{L}_{\text{cls}}^+(\mathcal{B}_p, \Theta, j, \epsilon)}{\partial \epsilon} \right| \right]. \quad (11)$$

Intuitively, a small τ_j means that a perturbation on coordinate j induces nearly identical changes at the receiver under both nominal and degraded conditions, making j a channel-resilient carrier for the backdoor injection. Moreover, a large ι_j indicates that perturbing latent coordinate j leads to a significant displacement of the classifier's decision boundaries. We now define modified and normalized scores:

$$\hat{\tau}_j = \frac{1/\tau_j}{\max_k (1/\tau_k)}, \quad \hat{\iota}_j = \frac{\iota_j}{\max_k \iota_k}.$$

The mask is defined as

$$m_j = \begin{cases} 1, & \text{if } \hat{\tau}_j + \hat{\iota}_j \text{ is among the top-}K, \\ 0, & \text{otherwise,} \end{cases} \quad (12)$$

enabling the selection of the most critical K dimensions of the latent space for the backdoor attack.

Stage two: Fine-tuning. At the second stage, we fine-tune the semantic encoder and the final classification head by using the poisoned dataset while keeping all other modules frozen. Note that this fine-tuning process excludes the latent trigger. The loss objective in the second stage is given by

$$\mathcal{L}_2 = (1 - \rho) \mathcal{L}_{\text{total}}(\mathcal{B}_c, \alpha) + \rho \mathcal{L}_{\text{total}}(\mathcal{B}_p, \alpha), \quad (13)$$

facilitating the injection of the backdoor into the semantic encoder while ensuring that the trigger effect persists through fine-tuning.

B. DSC-Adapted Baselines

We consider three baseline backdoor attacks [15]. We selected BadNets and RIPPLE, two popular yet compression-naive backdoor attacks, and implemented them using OpenBackdoor². BadNets inserts a fixed rare word trigger into a subset of training samples to induce a spurious association with a target label. RIPPLE follows the same trigger-insertion principle but applies it during fine-tuning of pre-trained models, thereby improving trigger persistence across adaptation phases. BadNetsDSC presents an adaptation of BadNets to the DSC setting, employing higher poisoning ratios and extended training to mitigate the effects of compression and channel noise. The training loss of this baseline is given by

$$\mathcal{L}_{\text{joint}}(\mathcal{B}; \Theta) = (1 - \rho) \mathcal{L}_{\text{total}}(\mathcal{B}_c; \Theta) + \rho \mathcal{L}_{\text{cls}}(\mathcal{B}_p; \Theta), \quad (14)$$

where $\mathcal{L}_{\text{total}}$ is defined in (6).

IV. EXPERIMENTAL RESULTS

We evaluate the proposed backdoor attack on the sentiment classification task³ trained with the SST-2 dataset⁴, assuming AWGN channel transmission with noise sampled from SNR in the range of [1, 10] dB and 16-QAM modulation. We average over five different seeds for all presented results. The semantic encoder uses RoBERTa [1], and the compression module adopts the hyperprior variational model from [7]. Unlike the original setup with a fixed channel encoder, we jointly train the semantic encoder, hyperprior module, and a learnable channel encoder with modulation.

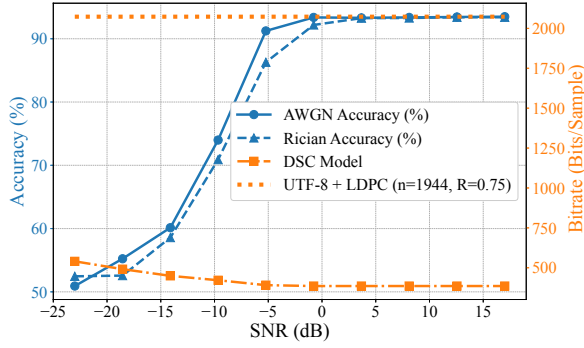


Fig. 2: Accuracy performance and bits per sample vs SNR.

Fig. 2 shows the classification performance before any backdoor is applied, demonstrating the classification accuracy and bitrate (bits/sample) versus SNR. It can be observed that the clean accuracy increases with the SNR and saturates around -6dB (AWGN) and 0dB (Rician), and that both curves follow nearly identical trends across the entire SNR range. Therefore, for brevity, we report results only for the AWGN channel in the following. The accuracy achieved by the DSC system reaches 94% while operating at under one-fifth the bit rate of UTF-8+LDPC (IEEE 802.11n). This extreme compression not only enhances communication efficiency but also inherently

suppresses localized token-level shifts, forming a natural first layer of defense against naive backdoor attacks. Table I reports

TABLE I: Performance with different mask sizes (SNR=3 dB)

K	8	16	32	64
Clean Accuracy (%)	93.4	94.1	93.4	93.5
Average ASR (%)	92.0	94.1	87.5	93.7
$\ \mathbf{m} \odot \beta_p\ / \ \mathbf{y}_p\ $ (%)	6.2	5.1	4.3	2.1

the average ASR and the classification accuracy on clean inputs under varying mask size K adopted in the first training stage. It is evident that the clean accuracy remains stable for all tested values. Thus, we select $K = 16$ for our remaining experiments. We compare the performance of BadNets,

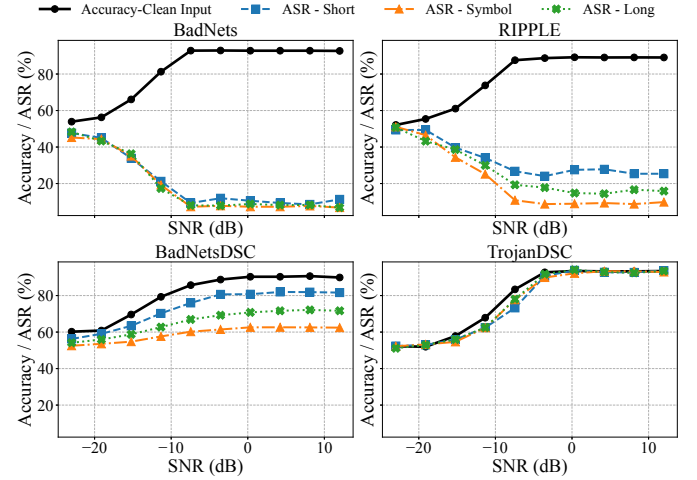


Fig. 3: Accuracy and ASR for different trigger types vs SNR.

RIPPLE, BadNetsDSC, and the proposed TrojanDSC across three trigger families—short tokens (cf, tq, mn, bb, mb), mathematical symbols ($\approx$, $\equiv$, $\in$, $\subseteq$, $\oplus$, $\otimes$), and long rare words (Psychotomimetic, Omphaloskepsis, Antidisestablishmentarianism, Xenotransplantation, Floccinaucinihilipilification) in Fig 3. All attack strategies use a 10% poison rate with the exception of BadNetsDSC, which uses 90%. For BadNets and RIPPLE, clean accuracy improves while ASR decreases as SNR increases. Specifically, BadNets' ASR reaches 11% for short tokens and falls below 7% for other token types, whereas RIPPLE decays more slowly at higher SNRs, settling near 25% for short tokens and 10 – 16% for symbols and long tokens. Counterintuitively, the decline in ASR as SNR increases occurs because, at low SNR, the decoder operates in a near-random regime and cannot reliably recover the waveform. As a result, predictions on poisoned inputs become effectively random, yielding a 50% ASR in our binary setting. As SNR improves and the signal quality increases, the backdoor becomes less likely to activate. BadNetsDSC has acceptable ASR at high SNRs (about 80% for short triggers, 70% for long words, and 65% for symbols) but its clean accuracy is about 89% at high SNRs. These results suggest that BadNetsDSC injects the backdoor into the DSC system more efficiently compared to BadNets and RIPPLE, but this requires a high poison rate of 90% and reduced classification performance on clean inputs. Fig. 3 also shows that TrojanDSC

²<https://github.com/thunlp/OpenBackdoor>

³Performance patterns for other tasks, such as toxicity detection and spam detection, are similar but omitted due to space limitations.

⁴<https://huggingface.co/datasets/stanfordnlp/sst2>

yields up to 94% ASR at an SNR of 3 dB, while causing negligible degradation of the accuracy on clean inputs. Indeed, its proper exploitation of stable and influential dimensions in the latent representation for backdoor injection ensures that the post-quantization latent remains on the desired side of the classification decision boundary. At 3 dB, our default rate setting yields an average of ≈ 460 bits/sample, similar to what is shown in Fig. 3, indicating that TrojanDSC’s backdoor remains effective under a realistic compression budget.

TABLE II: Ablation results for different poison rates ρ at an SNR of 3 dB

ρ	1%	5%	10%	20%	50%
hline TrojanDSC	80.8%	84.9%	94.2%	94.2%	94.3%
BadNetsDSC	24.2%	26.1%	11.5%	15.9%	17.1%

In Table II, we perform an ablation on the poison rate ρ at an SNR of 3 dB and report the ASR for TrojanDSC and the adapted BadNetsDSC baseline for a fair comparison. When evaluated at $\rho = 10\%$, BadNetsDSC achieves an ASR of 11.5% under the same TrojanDSC budget. In contrast, the ASR of TrojanDSC already saturates at $\rho = 10\%$, and further increasing the poison rate provides no additional gains. Fig. 4 demonstrates the clean accuracy and ASR versus SNR

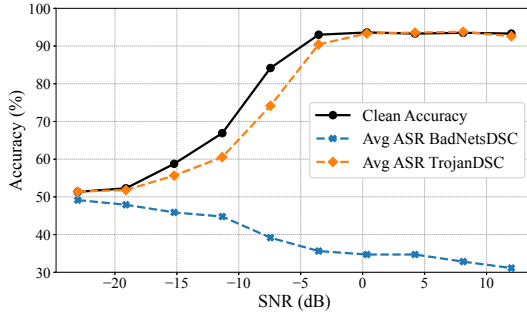


Fig. 4: Clean classification accuracy and ASR of BadNetsDSC and TrojanDSC vs SNR after fine-tuning defense

after the fine-tuning defense is applied to the TrojanDSC and BadNetsDSC backdoor models using the IMDB dataset⁵. We retrained both models until the clean accuracy is saturated. It can be observed that the clean accuracy remains above 94% for both backdoor methods at high SNRs. In addition, BadNetsDSC’s ASR decreases with the increasing SNR and falls below 40% at high SNRs, indicating that the catastrophic forgetting effect suppresses its trigger signals. In contrast, TrojanDSC maintains an approximate 93% ASR after the fine-tuning defense. This suggests that the latent-space trigger used in backdoor learning has effectively embedded a backdoor into the DSC model, making it harder to remove through fine-tuning. In other words, TrojanDSC is resilient to the fine-tuning defense. Given the risks posed by backdoor attacks, we recommend hardening the supply chain to prevent the embedding of backdoors in victim models and altering core physical-layer properties to disrupt the mapping from triggers to backdoor targets.

V. CONCLUSION

In this work, we proposed TrojanDSC, the novel backdoor attack strategy explicitly designed for modern DSC systems. Numerical studies showed that TrojanDSC achieves approximately 94% ASR at an SNR of 3 dB with negligible degradation in clean accuracy while remaining effective in fine-tuning. Our results further demonstrate that although DSC systems may resist conventional compression-naïve backdoor attacks, they remain vulnerable to advanced attacks like TrojanDSC, underscoring the urgent need for effective defense strategies. Backdoor attacks on other tasks, such as generation or question answering, and larger foundation models will be investigated in future works.

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⁵<https://huggingface.co/datasets/stanfordnlp/imdb>